COMPETENCY STANDARD

FOR

MOTOR CYCLE MAINTENANCE AND SERVICING

(TRANSPORT SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance,
Probashi Kallyan Bhaban (Level-15 & 16)
71-72 Old Elephant Road, Eskaton Garden
Dhaka-1000
Phone: +8802 55138753-55, Fax: +88 02 55138752
Website: https://sicip.gov.bd/

INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TRA-MCMS-01-G	Apply Occupational Health and Safety (OHS) Practice in the Workplace	1. Identify OHS policies and procedures. 2. Apply personal health and safety practices. 3. Report hazards and risks 4. Respond to emergencies.	10
SICIP-TRA-MCMS-02-G	Operate in a Team Environment	1 Identify team goals and work processes. 2 Identify own role and responsibilities within team. 3 Communicate and co-operate with team members. 4 Practice problem solving within the team.	10
		Total Hour	**20**

Sector Specific Competencies (40 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TRA-MCMS-01-S	Apply Green Practices	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
SICIP-TRA-MCMS-02-S	Work with Hand Tools and Power Tools	1. Identify and inspect hand and power tools 2. Use hand tools properly and safely 3. Operate power tools properly and safely. 4. Clean and maintain hand and power tools	30
		Total Hour	**40**

Occupation Specific Competencies (300 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-TRA-MCMS-01-O	Understand Motorcycle Systems and Components	1. Identify the key components of a motorcycle 2. Recognize motorcycle engines 3. Recognize motorcycle systems 4. Describe common motorcycle faults and their causes 5. Understand basic motorcycle maintenance	28
SICIP-TRA-MCMS-02-O	Perform Basic Motorcycle Inspections	1. Inspect the exterior of the motorcycle for visible damage 2. Check motor cycle wheel and tire 3. Check brakes, lights, and indicators 4. Check battery condition and charging system 5. Check suspension, steering, and wheel alignment	40
SICIP-TRA-MCMS-03-O	Service Engine on Motorcycles	1. Change engine oil and replace the oil filter 2. Inspect and replace spark plugs if necessary 3. Clean or replace air filters 4. Check and maintain fuel lines and systems 5. Test the engine after service 6. Adjust the idle speed and engine settings	72
SICIP-TRA-MCMS-04-O	Maintain Transmission System	1. Check the clutch system 2. Check the gearbox 3. Replace worn-out chain or sprockets 4. Test the transmission after repairs	60

Code	Unit of Competency	Elements of Competency	Duration (hours)
		5. Inspect and replace damaged seals and bearings	
SICIP-TRA-MCMS-05-O	Maintain and Adjust the Brake System of a Motorcycle	1. Inspect brake pads, discs, and fluid levels 2. Replace worn-out brake pads and adjust callipers 3. Perform bleed the brake lines and replace brake fluid 4. Test the brakes	60
SICIP-TRA-MCMS-06-O	Maintain the Electrical System of a Motorcycle	1. Test the battery voltage and charging capacity. 2. Inspect wiring. 3. Check the ignition system. 4. Replace faulty fuses and components. 5. Verify the Electrical system.	40
		Total Hours	**300 Hrs.**

COMPETENCY STANDARD: MOTOR CYCLE MAINTENANCE AND SERVICING

Generic Competencies

Unit of Competency: **APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-TRA-MCMS-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 **<u>OHS policies</u>** and safe operating procedures are interpreted 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Apply personal health and safety practices	2.1 OHS policies and procedures are interpreted in the workplace including **<u>personal protective equipment (PPE).</u>** 2.2 Common health issues are recognized. 2.3 Common safety issues are identified and applied.
3. Report hazards and risks	3.1 Hazards and risks are identified. 3.2 Hazards and risks assessment and controls are interpreted 3.3 Hazards and risk assessment are reported
4. Respond to emergencies	4.1 Respond to alarms and warning devices. 4.2 **<u>Emergency response plans and procedures</u>** are responded to. 4.3 **<u>First aid procedures</u>** during emergency situations are identified.

Range of variables:

Variables	**Range** (may include but not limited to)
1. OHS policies	1.1 Organizational OHS polices 1.2 International OHS requirements 1.3 Fire safety rules and regulations
2. Personal protective	2.1 Safety glasses

equipment	2.2 Ear plugs 2.3 Gloves 2.4 Apron 2.5 Helmet 2.6 Mask 2.7 Safety shoes
3. Emergency response plans and procedures	3.1 Firefighting procedures 3.2 Earthquake response procedures 3.3 Emergency response plans and procedures 3.4 Medical and first aid
4. First aid procedure	4.1 Washing of open wound 4.2 Washing chemically infected area 4.3 Applying bandage 4.4 Taking appropriate medicine

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace OHS policies and procedures 1.2 Work safety procedures 1.3 Emergency response procedures: 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 OHS awareness 1.6 Personal protective equipment (PPE) 1.7 Malfunctions and resolutions
2. Underpinning skill	2.1 Identifying OHS policies and procedures 2.2 Applying personal health and safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergencies
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace documents, signs and symbols 4.3 Codes of conduct 4.4 Projector 4.5 Learning manual

	4.6 Tools, equipment and facilities appropriate to the process or activities. 4.7 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified OHS policies and procedures 1.2 Applied personal health and safety practices (including PPE) 1.3 Reported hazards and risks 1.4 Responded to emergencies.
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **OPERATE IN A TEAM ENVIRONMENT**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-TRA-MCMS-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate in a team environment. It specifically includes the task of identifying team goals and work processes, identifying own role and responsibilities within team, communicating and cooperating with team members and practicing problem solving within the team.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Identify team goals and work processes	1.1. Roles and objectives of the team are identified and interpreted. 1.2. Roles and responsibilities of team members are identified and interpreted.
2. Identify own role and responsibilities within team	2.1 Personal role and responsibilities are identified within the team environment. 2.2 Reporting relationships are interpreted within team and external to team.
3. Communicate and co-operate with team members	3.1 Other teammates' tasks are identified and support provided when requested. 3.2 The team is encouraged through **sharing information** or expertise, working together to solve problems, and putting team success first. 3.3 Views and opinions of other team members are interpreted and respected.
4. Practice problem solving within the team	4.1 Problems faced at the individual and team level are identified and showed insight into the root causes of the problems. 4.2 A range of solutions and courses of action are identified together with benefits, costs, and risks associated with each. 4.3 The good ideas of others to help develop solutions are recognized and advice sought from those who have solved similar problems. 4.4 It is looked beyond the obvious and not stopped at the first answers.

Range of variables:

Variables	**Range** (may include but not limited to)

1. Sharing information	1.1. Agenda 1.2. Minutes 1.3. progress and incident reports 1.4. Operational manuals 1.5. Visual and graphic materials 1.6. Emails and SMS 1.7. Phone directory 1.8. Policy, procedure and standards 1.9. OHS information

Curricular Content Guide

1. Underpinning knowledge	1.1. Team goals and work processes 1.2. Roles and responsibilities of team members 1.3. Personal role and responsibilities 1.4. Sharing information 1.5. Views and opinions 1.6. Problem solving within the team
2. Underpinning skill	2.1. Identifying team goals and work processes 2.2. Identifying own role and responsibilities within team 2.3. Communicating and cooperating with team members 2.4. Practicing problem solving within the team
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Eagerness to learn 3.5 Tidiness and timeliness 3.6 Environmental concerns 3.7 Respect for rights of peers and seniors at workplace 3.8 Communication with peers and seniors at workplace
4. ‘Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified team goals and work processes 1.2 identified own role and responsibilities within team 1.3 communicated and cooperated with team members 1.4 practiced problem solving within the team
2. Methods of assessment	Competency should be assessed by:

	2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: **APPLY GREEN PRACTICES**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-TRA-MCMS-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply green practices. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 **<u>Principles of green practices</u>** are explained. 1.2 **<u>Sources of environmental impacts</u>** during construction activities are identified. 1.3 Work activities contributing to environmental degradation, improper disposal of materials and excessive energy use are explained. 1.4 **<u>Ways of mitigating environmental impacts</u>** in construction sector are explained.
2. Minimize resource use in the workplace	2.1 Water, energy, and raw material consumption are documented. 2.2 **<u>Recyclable and non-recyclable items</u>** are identified. 2.3 Procedures to reduce resource consumption are implemented. 2.4 Sustainable alternatives to fossil-based energy resources are explored and applied.
3. Implement waste management practices	3.1 **<u>Different types of waste</u>** are identified. 3.2 Hazardous waste is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	**Range** May include but not limited to:
1. Principles of green practices	1.1 Reducing energy consumption 1.2 6R for waste management 1.2.1 Refuse 1.2.2 Reduce 1.2.3 Reuse 1.2.4 Recycle 1.2.5 Recover 1.2.6 Repair

	1.3 use of sustainable materials with low environmental impact 1.4 Recycling materials 1.5 Sustainable transportation
2. Sources of environmental impacts	2.1 Air pollution 2.1.1 Dust generation 2.1.2 Emission from machinery and vehicles 2.2 Water pollution 2.2.1 Debris and sediments 2.2.2 Chemical's leakage 2.3 Noise pollution 2.4 Waste generation 2.4.1 waste 2.4.2 Non-recyclable and hazardous materials 2.5 Resource Depletion 2.5.1 Excessive use of raw materials 2.5.2 Non-renewable material uses like use of fossil fuel, steel cement etc.
3. Ways of mitigating environmental impacts	3.1 Utilizing energy-efficient equipment 3.2 Adopting renewable energy sources 3.3 Implementing site protection measures 3.4 Using reusable materials 3.5 Choosing sustainable materials 3.6 Using noise-reducing equipment and scheduling work appropriately 3.7 Optimizing logistics and delivery
4. Recyclable and non-recyclable items	4.1 Recyclable items 4.1.1 Metal 4.1.2 Wood 4.1.3 Brick 4.1.4 Concrete 4.2 Non-recyclable items 4.2.1 Paints and coatings 4.2.2 Contaminated materials like lead paint, asbestos 4.2.3 Treated wood
5. Different types of waste	5.1 Wood waste 5.2 Metal waste 5.3 Plastic waste 5.4 Asbestos waste 5.5 Gypsum board waste 5.6 Packaging waste 5.7 Organic waste 5.8 Chemical West 5.9 Fuel/Oil west

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in shipbuilding sector 1.2 Key terms and symbols related to environmental sustainability in construction drawings 1.3 Sources of environmental impacts in construction 1.4 Methods to minimize resource consumption (water, energy, raw materials) 1.5 Waste management practices 1.6 Sustainable alternatives to fossil-based energy resources 1.7 Personal and workplace habits for reducing environmental impact
2. Underpinning Skills	2.1 Identifying environmental impacts in construction activities 2.2 Applying methods to minimize resource use in the workplace 2.3 Implementing waste management practices 2.4 Following procedures to safely dispose of hazardous materials 2.5 Documenting water, energy, and material consumption in the workplace 2.6 Performing green practices in personal and professional activities
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of construction hand tools and power tools 4.3 Safety gear 4.4 Pens 4.5 Papers 4.6 Work books 4.7 Operation and maintenance manuals 4.8 Waster segregation bins

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Explained the principles of green practices in construction sector 1.2 Identified sources of environmental impacts 1.3 Implemented procedures to minimize resource use (water, energy, raw materials) 1.4 Disposed of hazardous waste in compliance with safety and environmental standards 1.5 Practiced green habits to reduce waste in both personal and professional life
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **WORK WITH HAND AND POWER TOOLS**	**Nominal Duration:** 30 hrs.	**Unit Code:** SICIP-TRA-MCMS-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to use hand and power tools in the workplace. It specifically includes the task of identifying and inspecting hand and power tools, using hand tools properly and safely, operating power tools properly and safely and cleaning and maintaining hand and power tools.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and inspect hand and power tools	1.1 Appropriate hand and power tools are identified. 1.2 Application of hand and power tools is recognized. 1.3 Usability of hand and power tools is checked and verified.
2. Use hand tools properly and safely	2.1. Appropriate **<u>hand tools</u>** are selected. 2.2. Safety precautions are ensured before using hand tools. 2.3. Unsafe or faulty hand tools are identified and marked for repair. 2.4. **<u>Measuring tools</u>** are checked and calibrated before use. 2.5. Use hand tools properly and safely to perform work activity.
3. Operate power tools properly and safely	3.1. Appropriate **<u>power tools</u>** are selected. 3.2. Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.3. Safety precautions are ensured before using power tools in accordance with manufacturer's operating specification. 3.4. Proper sequence of operation applied for using power tools. 3.5. Unsafe or faulty power tools are identified and marked for repair. 3.6. Operate power tools properly and safely to perform work activity.
4. Clean and maintain hand and power tools	4.1. Dust and foreign matter is removed from hand and power tools in accordance to workplace standards. 4.2. Condition of hand and power tools is checked after use and reported. 4.3. Appropriate lubricant is applied after use and prior to storage. 4.4. Defective hand and power tools are inspected and

	repaired or replaced. 4.5. Hand and power tools are stored and secured in accordance with workplace requirements.

Range of Variables

Variable	**Range** May include but not limited to:
1. Hand tools	1.1 Hammer 1.2 Bench vice 1.3 Files 1.4 Wrenches 1.5 Pliers 1.6 Scriber 1.7 Scraper Screw 1.8 Surface plate 1.9 Gauge 1.10 Tap sets 1.11 Die sets 1.12 Hacksaw 1.13 Spanners 1.14 Vice grip 1.15 Wire cutters 1.16 Clamps 1.17 Jacks
2. Power tools	2.1 Drills 2.2 Rivet gun 2.3 Grinders 2.4 Pneumatic wrenches 2.5 Press machine 2.6 Cutting 2.7 Saws 2.8 Soldering iron
3. Measuring tools	3.1 Micrometer 3.2 Megger 3.3 Measuring tape 3.4 Hose level 3.5 Water level 3.6 Calipers 3.7 Steel rule 3.8 Meter rule 3.9 Spirit level 3.10 Protractor 3.11 Tri-square 3.12 Gauges

Curricular Content Guide

1. Underpinning Knowledge	1.1 Information on hand and power tools, their functions and use 1.2 Procedures for safely using hand and power tools
2. Underpinning Skills	2.1 Identifying hand, power and measuring tools 2.2 Following safety precautions when using hand, power and measuring tools 2.3 Using hand and measuring tools correctly and safely in accordance with manufacturer's operating specification 2.4 Operating power tools correctly and safely in accordance with manufacturer's operating specification 2.5 Cleaning and maintaining hand and power tools after use 2.6 Applying appropriate lubricant on hand and power tools after use and prior to storing
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and selected appropriate hand and power tools for work to be performed 1.2 identified and used measuring and testing tools appropriate to work activity 1.3 followed safety precautions when using hand and power tools 1.4 operated power tools safely and pursuant to manufacturer's operating specification

	1.5 performed cleaning and maintenance of hand and power tools after use and prior to storing
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: UNDERSTAND MOTORCYCLE SYSTEMS AND COMPONENTS	Nominal Duration: 28 Hrs.	Unit Code: SICIP-TRA-MCMS-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to understand motorcycle systems and components. It specifically includes the task of identifying the key components of a motorcycle, recognizing motorcycle engines, recognizing motorcycle systems, describing common motorcycle faults and their causes and understanding basic motorcycle maintenance.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify the key components of a motorcycle	1.1 Frame is identified, and its role in providing structural support for all motorcycle components is understood. 1.2 Fuel tank is identified, and its function in storing and supplying fuel to the engine is recognized. 1.3 **Battery** is identified, and its role in supplying electrical power to the motorcycle is understood. 1.4 **Tires** are identified, and their function in providing traction, stability, and shock absorption is recognized. 1.5 **Wheels** are identified, and their role in supporting the tires and enabling rotation is understood. 1.6 **Brakes** are identified, and their function in providing stopping power through friction is recognized.
2. Recognize motorcycle engines.	2.1 **Motorcycle engines** are recognized based on their type. 2.2 Engine displacement is identified. 2.3 **Engine configuration** is recognized and understood. 2.4 **Engine's cooling method** is identified. 2.5 Engine's position is recognized and categorized. 2.6 **Common engine components** are identified for their specific roles in the engine operation.
3. Recognize motorcycle systems.	3.1 **Electrical system** is identified. 3.2 Fuel system is recognized for supplying fuel to the engine. 3.3 Braking system is identified for ensuring stopping power. 3.4 Cooling system is recognized for maintaining optimal engine temperature. 3.5 Suspension system is identified for providing a smooth ride. 3.6 Transmission system is identified for transferring engine power to the wheels.

4. Describe common motorcycle faults and their causes.	4.1 Faults and causes in the engine are identified. 4.2 Transmission faults and causes are recognized. 4.3 Electrical faults and causes are described. 4.4 Fuel system faults and causes are identified. 4.5 Braking system faults and causes are noted. 4.6 Suspension faults and causes are recognized. 4.7 Cooling system faults and causes are described. 4.8 Exhaust system faults and causes are identified.
5. Understand basic motorcycle maintenance.	5.1 Routine maintenance tasks are understood and recognized as essential for the motorcycle's longevity. 5.2 Proper wheel and tire maintenance is understood to maintain safety and handling. 5.3 Role of the battery and electrical system is understood. 5.4 Importance of maintaining the braking system is recognized to ensure safety. 5.5 Significance of maintaining the suspension system is understood to ensure a smooth ride. 5.6 Regular inspections of the clutch, transmission, and exhaust system are understood.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Battery	1.1 Wet Cell or flooded cell batteries 1.2 Dry Cell sealed type batteries 1.3 Gel Batteries
2. Tires	2.1 Sport Tires – High grip, for aggressive riding. 2.2 Cruiser Tires – Comfortable, long-lasting for cruisers. 2.3 Touring Tires – Stable, comfortable for long-distance. 2.4 Off-Road Tires – Deep treads, for dirt and trails. 2.5 Dual-Sport Tires – Mix of on-road and off-road performance. 2.6 Bias Ply Tires – Durable, but less responsive at high speeds. 2.7 Radial Ply Tires – Flexible, smooth, for better handling. 2.8 Racing Tires – Super grip, track-specific.
3. Wheels	3.1 Spoke Wheels – Classic, strong, lightweight. 3.2 Cast Alloy Wheels – Durable, low maintenance. 3.3 Forged Wheels – Lighter, stronger, high-performance. 3.4 Mag Wheels – Super light, high-performance. 3.5 Wire Wheels – Vintage look, intricate design. 3.6 Composites – Carbon fiber, ultra-light, expensive. 3.7 Tubeless Wheels – No inner tube, safer, easy repairs. 3.8 Dual-Tire Wheels – For dual-sport bikes, on/off-road.
4. Brakes	4.1 Disc Brakes 4.1.1 Hydraulic Brakes

	4.1.2 Mechanical Brakes 4.2 Drum Brakes 4.2.1 Mechanical Brakes 4.3 ABS (Anti-lock Braking System) 4.4 CBS (Combined Braking System)
5. Motorcycle engines	5.1 4-stroke engine 5.2 2-stroke engine
6. Engine configuration	6.1 Single-Cylinder 6.2 Parallel-Twin 6.3 V-Twin 6.4 Inline-Three 6.5 Inline-Four 6.6 V-Four
7. Engine's cooling method	7.1 Water cooling 7.2 Air cooling
8. Common engine components	8.1 Cylinder 8.2 Piston 8.3 Crankshaft 8.4 Valves 8.5 Camshaft 8.6 Cylinder Head 8.7 Cylinder block 8.8 Tapet cover 8.9 Spark Plug 8.10 Carburetor/Fuel Injector 8.11 Timing Chain/Belt 8.12 Oil Pump 8.13 Self-starter 8.14 Clutch
9. Electrical system	9.1 Battery 9.2 Alternator 9.3 Regulator/Rectifier 9.4 Ignition Coil 9.5 Spark Plug 9.6 Starter Motor 9.7 Dashboard 9.8 Starting switch 9.9 Stator 9.10 Fuse Box 9.11 Headlight/Taillight 9.12 Turn Signals 9.13 Wiring Harness 9.14 Voltage Regulator

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify the main structural components of a motorcycle. 1.2 Understand motorcycle engine fundamentals. 1.3 Concept of power transmission system. 1.4 Identify key auxiliary systems and explain their specific functions. 1.5 Interpret common faults and their causes. 1.6 Understand the role of the electrical system. 1.7 Concept of braking systems and their importance for safety through effective stopping power. 1.8 Understand suspension systems and their role in stability, comfort, and handling. 1.9 Identify and apply principles of regular maintenance practices. 1.10 Explain the importance of preventive maintenance in extending motorcycle life.
2. Underpinning Skills	2.1. Detecting faults and causes in engine, transmission, electrical, fuel, braking, suspension, cooling, and exhaust systems. 2.2. Performing routine maintenance tasks to ensure motorcycle longevity. 2.3. Maintaining wheels and tires for safety and handling. 2.4. Maintaining the battery and electrical system for reliable performance. 2.5. Maintaining the braking system to ensure safety. 2.6. Maintaining the suspension system for smooth riding. 2.7. Inspecting clutch, transmission, and exhaust system regularly.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified the key components of a motorcycle

	1.2 recognized motorcycle engines 1.3 recognized motorcycle systems 1.4 described common motorcycle faults and their causes 1.5 understood basic motorcycle maintenance
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM BASIC MOTORCYCLE INSPECTIONS**	**Nominal Duration:** 40 Hrs.	**Unit Code:** SICIP-TRA-MCMS-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform basic motorcycle inspections. It specifically includes the task of inspecting the exterior of the motorcycle for visible damage, checking motorcycle wheel and tire, checking brakes, lights and indicators, checking battery condition and charging system and checking suspension, steering and wheel alignment.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Inspect the exterior of the motorcycle for visible damage.	1.1 Condition of the wheels and tires is checked to ensure there are no punctures, cracks, or excessive wear. 1.2 Mirrors, lights, and indicators are examined for any cracks, damage, or malfunctioning. 1.3 Exhaust system is inspected for any visible holes, rust, or damage. 1.4 Seat and handlebars are checked for signs of wear or damage that could impact comfort and control. 1.5 **<u>Suspension components</u>** are visually inspected for any leaks, cracks, or damage that could affect handling. 1.6 Any visible fluid leaks are identified and noted for further investigation.
2. Check motor cycle wheel and tire.	2.1 Wheels are inspected for visible damage. 2.2 Tires are checked for proper inflation. 2.3 Tire tread is examined for excessive wear or uneven wear patterns. 2.4 Tire sidewalls are inspected to potential tire failure. 2.5 Wheel spokes (if applicable) are checked for tightness. 2.6 Tires are assessed for proper alignment with the wheel to ensure smooth rotation and stability. 2.7 Any damage or wear found on the wheels or tires is documented.
3. Check brakes, lights, and indicators.	3.1 Brakes are checked for proper functionality to ensure efficient braking performance. 3.2 Brake lever or pedal is tested to ensure it functions smoothly and engages the brakes without delay. 3.3 **<u>Lights</u>** and brake lights, are checked to confirm they are working correctly. 3.4 Indicators are tested to ensure they are functioning properly. 3.5 Wiring and connections for brakes, lights, and indicators are visually inspected.

	3.6 Any malfunctioning or damaged brakes, lights, or indicators are documented.
4. Check battery condition and charging system.	4.1 **<u>Battery</u>** is checked by inspecting the charge level. 4.2 Battery terminals are inspected for corrosion or loose connections. 4.3 Battery is tested to ensure it holds a charge and does not show signs of damage. 4.4 Charging system is checked to ensure it is properly charging the battery while the engine is running. 4.5 Voltage output from the charging system is tested to ensure it is within the specified range for the battery. 4.6 Any issues with the battery or charging system are documented.
5. Check suspension, steering, and wheel alignment.	5.1 **<u>Suspension system</u>** is checked for proper function. 5.2 Steering system is inspected to ensure smooth operation. 5.3 Steering head bearings are checked for smooth rotation and proper tightness. 5.4 Wheel alignment is examined to ensure the wheels are properly aligned. 5.5 Front and rear suspension settings are assessed and adjustments are made if necessary. 5.6 Any issues with the suspension, steering, or wheel alignment are documented.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Suspension components	1.1 Forks (Front Suspension) 1.2 Shock Absorbers (Rear Suspension) 1.3 Springs 1.4 Damping Mechanism 1.5 Swingarm
2. Lights	2.1 Headlight 2.2 Taillight 2.3 Turn Signals 2.4 Brake Light 2.5 Indicators 2.6 Daytime Running Lights (DRL) 2.7 Fog Lights
3. Battery	3.1 Lead acid 3.2 AGM 3.3 EFB 3.4 Lithium
4. Suspension system	4.1 **Front Suspension:** 4.1.1 Telescopic Fork (Conventional) 4.1.2 Upside Down (USD) Fork

	4.1.3 Springer Suspension 4.1.4 Girder Fork 4.1.5 Leading Link Suspension 4.2 **Rear Suspension:** 4.2.1 Twin Shock Absorber 4.2.2 Mono-shock Suspension 4.2.3 Swing Arm Suspension 4.2.4 Cantilever Suspension

Curricular Content Guide

1. Underpinning Knowledge	1.1. Concept of motorcycle pre-ride inspection and its importance for safety, performance, and maintenance. 1.2. Identify wheel and tire defects and interpret their impact on handling and stability. 1.3. Understand the function and inspection methods for mirrors, lights, indicators, and wiring connections. 1.4. Interpret the role of the exhaust system and recognize visible faults. 1.5. Understand seat, handlebar, and suspension inspection for comfort, stability, and safe control of the motorcycle. 1.6. Identify brake system components and interpret signs of wear, malfunction, or reduced performance. 1.7. Understand the function of the battery and charging system. 1.8. Interpret suspension system checks. 1.9. Understand the importance of documenting inspection results, faults, and maintenance needs.
2. Underpinning Skills	2.1. Checking wheels and tires for punctures, cracks, wear, and proper inflation. 2.2. Inspecting mirrors, lights, indicators, and wiring for damage or malfunction. 2.3. Examining the exhaust system for holes, rust, or visible damage. 2.4. Checking seat, handlebars, and suspension components for wear, leaks, or cracks. 2.5. Inspecting tire tread, sidewalls, alignment, and spoke tightness. 2.6. Testing brakes, levers, and pedals for smooth functionality. 2.7. Checking brake lights, indicators, and connections for proper operation. 2.8. Inspecting and testing the battery for charge level, corrosion, and functionality. 2.9. Checking charging system and voltage output while engine is running. 2.10. Inspecting steering system, bearings, and wheel

	alignment.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 inspected the exterior of the motorcycle for visible damage 1.2 checked motor cycle wheel and tire 1.3 checked brakes, lights, and indicators 1.4 checked battery condition and charging system 1.5 checked suspension, steering, and wheel alignment
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **SERVICE ENGINE ON MOTORCYCLES**	Nominal Duration: 72 Hrs.	Unit Code: SICIP-TRA-MCMS-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to service engine on motorcycles. It specifically includes the tasks of changing engine oil and replace the oil filter, inspecting and replacing spark plugs if necessary, cleaning and maintaining fuel lines and systems, testing the engine after service and adjusting the idle speed and engine settings.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Change engine oil and replace the oil filter	1.1 Oil drain plug is removed and the old engine oil is drained into an appropriate container. 1.2 Old oil filter is removed carefully, ensuring that any residual oil is contained and disposed of properly. 1.3 New oil filter is installed, ensuring it is properly tightened and sealed to prevent leaks. 1.4 Oil drain plug is reinstalled and tightened to prevent oil leaks. 1.5 Correct **types of engine oil** and appropriate amount are added, following the manufacturer's specifications. 1.6 Oil level is checked after adding the new oil, and adjustments are made. 1.7 Engine is started and allowed to run for a few minutes, and the oil level is rechecked.
2. Inspect and replace spark plugs if necessary.	2.1 **Spark plugs** are inspected for wear, corrosion, or damage. 2.2 Spark plug gap is checked using a feeler gauge. 2.3 Condition of the spark plugs electrode is examined. 2.4 New spark plugs, matching the specifications for the motorcycle are installed if required. 2.5 Spark plugs are checked again after installation. 2.6 Motorcycle is started to verify that the engine is running smoothly, and the spark plug function is confirmed. 2.7 Used spark plugs are disposed of properly according to local regulations.
3. Clean or replace air filters	3.1 Air filter is inspected for dirt, debris, and damage, ensuring it is clean and functional. 3.2 If the air filter is reusable, it is carefully removed and cleaned using compressed air to remove dirt and dust. 3.3 If the air filter is excessively dirty, torn, or damaged, it is replaced. 3.4 Cleaned or new air filter is installed.

	3.5 Air filter housing is checked for any dirt or debris, and it is cleaned before the filter is reinstalled. 3.6 Motorcycle is started to verify that the engine is running smoothly, ensuring proper air filtration.
4. Check and maintain fuel lines and systems.	4.1 Fuel filter is checked for any clogging or contamination and is replaced if necessary. 4.2 **Fuel system components** are inspected for proper function and cleanliness. 4.3 Fuel tank is checked for secure attachment and inspected for rust or corrosion. 4.4 Fuel lines are cleaned, if required, to remove any debris or buildup that could restrict fuel flow. 4.5 **Fuel system** is tested for pressure to ensure it is within the manufacturer's specifications. 4.6 Fuel lines and system are reassembled, and the motorcycle is started.
5. Test the engine after service.	5.1 Engine is started and allowed to idle to ensure smooth operation and proper engine performance. 5.2 Engine is checked for any unusual noises, vibrations, or irregularities that may indicate a problem. 5.3 Oil pressure, temperature, and other gauges are monitored. 5.4 Engine is tested under load by gradually increasing speed. 5.5 **Exhaust system** is checked for proper emissions and any signs of leaks. 5.6 Fuel system is inspected during testing. 5.7 Throttle and clutch operation are checked for smoothness and responsiveness.
6. Adjust the idle speed and engine settings	6.1 Idle speed is adjusted by turning the idle screw to the manufacturer's recommended RPM. 6.2 **Engine settings** are checked and adjusted to ensure optimal performance and fuel efficiency. 6.3 Throttle response is tested to ensure it operates smoothly without hesitation or stalling. 6.4 Engine is allowed to idle after adjustments, and the idle speed is rechecked. 6.5 Ignition timing is checked and adjusted, if necessary. 6.6 Engine is tested under load after adjustments.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Types of engine oil	1.1 Mineral Oil – 20W-50 1.2 Semi-Synthetic Oil – 10W-40 1.3 Full Synthetic Oil – 5W-40
2. Spark plugs	2.1 Single electrode 2.2 Double electrode

3. Fuel system components	3.1 Fuel Tank 3.2 Fuel Pump 3.3 Fuel Filter 3.4 Fuel Lines 3.5 Fuel Injectors 3.6 Carburetor 3.7 Electronics Control Units (ECU) 3.8 Fuel Pressure Regulator 3.9 Fuel Rail 3.10 Fuel Sending Unit 3.11 Fuel Cap 3.12 Throttle Body
4. Fuel system	4.1 Carburetor Fuel System 4.2 Fuel Injection System 4.3 Electronic Fuel Injection (EFI) System
5. Exhaust system	5.1 Single Exhaust System 5.2 Dual Exhaust System 5.3 2-in-1 Exhaust System 5.4 2-in-2 Exhaust System
6. Engine settings	6.1 Idle speed (1000–1500 RPM) 6.2 Air–fuel mixture 6.3 Ignition timing 6.4 Throttle free play 6.5 Valve timing and clearance 6.6 Carburetor/Fuel injection setting

Curricular Content Guide

1. Underpinning Knowledge	1.1. Concept of engine lubrication, oil circulation. 1.2. Identify oil filter and drain plug components 1.3. Understand correct procedures for draining, replacing, and refilling engine oil. 1.4. Understand spark plug functions, inspection methods and correct replacement procedures. 1.5. Interpret the role of the air filter in combustion efficiency. 1.6. Identify fuel system components and understand their role in fuel delivery and combustion. 1.7. Understand inspection, cleaning, and testing methods. 1.8. Interpret gauges and indicators for monitoring engine performance and identifying irregularities. 1.9. Understand throttle, clutch, and idle speed adjustments. 1.10. Understand final performance testing methods.
2. Underpinning Skills	2.1 Draining old engine oil and removing the oil filter safely. 2.2 Installing new oil filter and tightening the oil drain plug securely.

	2.3 Adding correct engine oil, checking levels, and making adjustments. 2.4 Installing new spark plugs as per specifications and confirming performance. 2.5 Disposing of used spark plugs properly. 2.6 Inspecting, cleaning, or replacing the air filter as required. 2.7 Inspecting fuel tank, lines, and components for rust, leaks, or debris. 2.8 Checking exhaust system for emissions and leaks. 2.9 Testing throttle and clutch for smooth operation. 2.10 Checking and adjusting ignition timing if necessary. 2.11 Testing engine response and performance under load.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 changed engine oil and replace the oil filter 1.2 inspected and replaced spark plugs if necessary 1.3 cleaned or replaced air filters 1.4 checked and maintained fuel lines and systems 1.5 tested the engine after service 1.6 adjusted the idle speed and engine settings
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **MAINTAIN TRANSMISSION SYSTEM**	**Nominal Duration:** 60 Hrs.	**Unit Code:** SICIP-TRA-MCMS-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to maintain transmission system. It specifically includes the tasks of checking the clutch system, checking the gearbox, replacing worn-out chain or sprockets, testing the transmission after repairs, and inspecting and replacing damaged seals and bearings.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Check the clutch system	1.1 Clutch lever and cable are inspected to ensure smooth operation without excessive play. 1.2 Clutch plates are checked for wear, damage, or signs of overheating. 1.3 Clutch spring tension is measured and compared with the manufacturer's specifications. 1.4 Clutch fluid level is inspected and replenished if required. 1.5 Clutch engagement and disengagement are tested to confirm proper functioning. 1.6 **Faults or abnormalities in the clutch system** are recorded and reported according to workplace procedures.
2. Check the gearbox	2.1 Gearbox housing is visually inspected for cracks, leaks, or loose fasteners. 2.2 Oil level and condition are checked, and contamination or shortage is identified. 2.3 Gear shifting is tested while the engine is running, and abnormal noise or resistance is noted. 2.4 Each gear is engaged and disengaged to ensure smooth operation and correct alignment. 2.5 Any unusual vibration, dragging, or slippage is observed during gear changes.
3. Replace worn-out chain or sprockets	3.1 Drive chain and sprockets are inspected and wear or damage is identified. 3.2 Chain tension is released and the worn-out chain or sprockets are removed using proper tools. 3.3 New chain and sprockets are installed according to manufacturer's specifications. 3.4 Chain alignment and correct tension are adjusted and ensured. 3.5 Fasteners are tightened to the recommended torque settings.

	3.6 Motorcycle is test-run, and smooth power transmission without noise or vibration is confirmed.
4. Test the transmission after repairs.	4.1 Repaired transmission components are visually checked for correct installation. 4.2 Lubricant level and condition are verified before testing. 4.3 Engine is started and the gearbox is operated through all gears. 4.4 Smooth engagement and disengagement of gears are observed without abnormal noise. 4.5 Motorcycle is test-ridden under controlled conditions to confirm proper functioning.
5. Inspect and replace damaged seals and bearings.	5.1 **Transmission seals and bearings** are visually inspected for wear, leakage, cracks, or play. 5.2 Damaged or worn seals and bearings are carefully removed using proper tools. 5.3 Replacement parts are cleaned, lubricated, and checked for specification before fitting. 5.4 New seals and bearings are installed according to manufacturer's instructions. 5.5 All fasteners and retaining clips are tightened to recommended torque settings. 5.6 Transmission is re-assembled and tested to confirm smooth, noise-free operation.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Faults or abnormalities in the clutch system	1.1 Clutch slipping 1.2 Clutch drag 1.3 Hard/heavy clutch lever 1.4 Clutch judder/shudder 1.5 Clutch noise 1.6 Clutch overheating/burning smell 1.7 Worn clutch plates 1.8 Weak or broken clutch springs 1.9 Damaged or stretched clutch cable 1.10 Oil contamination on plates
2. Transmission seals and bearings	**Transmission Seals** 2.1 Input shaft oil seal 2.2 Output shaft oil seal 2.3 Gearshift lever seal 2.4 Clutch pushrod seal 2.5 Countershaft sprocket seal **Transmission Bearings** 2.6 Input shaft bearing 2.7 Output shaft bearing 2.8 Main shaft bearing

	2.9 Countershaft bearing 2.10 Needle roller bearing 2.11 Ball bearing 2.12 Thrust bearing

Curricular Content Guide

1. Underpinning Knowledge	1.1. Concept of a clutch system is understood. 1.2. Identify common clutch components and their functions. 1.3. Interpret manufacturer's specifications for clutch spring tension, fluid levels, and wear limits. 1.4. Understand causes of clutch problems like slippage, dragging, overheating, or abnormal noise. 1.5. Concept of gearbox operation is explained. 1.6. Identify gearbox components. 1.7. Understand the importance of proper lubrication in clutch and gearbox systems. 1.8. Interpret signs of gearbox faults such as vibration, resistance, oil leaks, or abnormal noise. 1.9. Concept of drive chain and sprockets is understood. 1.10. Identify wear and alignment issues in chains and sprockets that affect performance and safety.
2. Underpinning Skills	2.1. Inspecting clutch lever, cable, plates, springs, and fluid level. 2.2. Testing clutch engagement and disengagement for smooth operation. 2.3. Inspecting gearbox housing for cracks, leaks, and loose fasteners. 2.4. Testing gear shifting and engaging all gears for smooth alignment. 2.5. Observing noise, vibration, dragging, or slippage during gear changes. 2.6. Tightening fasteners to recommended torque settings. 2.7. Test-running the motorcycle for smooth power transmission. 2.8. Inspecting and replacing worn seals and bearings using proper tools. 2.9. Reassembling transmission and testing for smooth, noise-free operation.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace

	3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 checked the clutch system 1.2 checked the gearbox 1.3 replaced worn-out chain or sprockets 1.4 tested the transmission after repairs 1.5 inspected and replaced damaged seals and bearings
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **MAINTAIN AND ADJUST THE BRAKE SYSTEM OF A MOTORCYCLE**	**Nominal Duration:** 60 Hrs**.**	**Unit Code:** SICIP-TRA-MCMS-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to maintain and adjust the brake system of a motorcycle. It specifically includes the tasks of inspecting brake pads, discs and fluid levels, replacing worn-out brake pads and adjust callipers, performing bleed the brake lines and replace brake fluid and testing the brakes.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Inspect brake pads, discs, and fluid levels	1.1 **<u>Brake pads</u>** are inspected for correct thickness, cracks, or uneven wear. 1.2 Brake discs are examined for warping, cracks, and surface damage. 1.3 Brake fluid and level is checked against the recommended marks in the reservoir. 1.4 Condition of **<u>brake fluid</u>** is observed for discoloration or contamination. 1.5 Brake lines and hoses are inspected for leakage, cracks, or looseness.
2. Replace worn-out brake pads and adjust callipers	2.1 **<u>Brake caliper</u>** assembly is removed carefully using proper tools. 2.2 Worn-out brake pads are taken out and inspected for uneven wear or damage. 2.3 New brake pads of correct specification are installed in the caliper. 2.4 Caliper pistons are checked, cleaned, and reset to proper position. 2.5 Brake caliper is reinstalled, and all fasteners are tightened to the specified torque. 2.6 Brake calipers are adjusted to ensure correct pad alignment and smooth operation. 2.7 Brake lever/pedal is operated to confirm proper pad contact with the disc.
3. Perform bleed the brake lines and replace brake fluid.	3.1 Brake fluid reservoir is opened and the old brake fluid is drained properly. 3.2 Bleed screws on the calipers are loosened and brake lines are bled to remove air. 3.3 New brake fluid of the specified grade is filled into the reservoir. 3.4 Brake lever/pedal is pumped until clear, bubble-free fluid flows from the bleed valve. 3.5 Bleed screws are tightened securely and the reservoir

	cap is refitted. 3.6 Brake lever/pedal free play and hydraulic pressure are tested to ensure proper function. 3.7 All fluid leaks are checked and cleaned, and the service is documented.
4. Test the brakes.	4.1 **Brake lever** and **brake pedal** are pressed to test hydraulic pressure and free play. 4.2 Wheel rotation is checked to confirm proper engagement and release of the brakes. 4.3 Front and rear brakes are applied separately to ensure balanced stopping force. 4.4 Abnormal noise, vibration, or brake drag is observed during testing. 4.5 A controlled road test is carried out to verify braking efficiency under normal conditions.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Brake pads	1.1 Organic (Non-Asbestos Organic – NAO) brake pads 1.2 Semi-metallic brake pads 1.3 Sintered (metallic) brake pads 1.4 Ceramic brake pads
2. Brake fluid	2.1 DOT 3 Brake Fluid 2.2 DOT 4 Brake Fluid 2.3 DOT 5 Brake Fluid (silicone-based) 2.4 DOT 5.1 Brake Fluid (glycol-based)
3. Brake caliper	3.1 Caliper body 3.2 Caliper piston(s) 3.3 Dust seal 3.4 Oil seal (piston seal) 3.5 Bleeder screw (bleed nipple) 3.6 Caliper mounting bolts 3.7 Pad retaining pin/clip 3.8 Anti-rattle clip (spring clip) 3.9 Guide pin / Slide pin (for floating calipers)
4. Brake lever	4.1 Brake lever (handle) 4.2 Pivot bolt & nut 4.3 Return spring 4.4 Adjusting screw (free play adjuster) 4.5 Master cylinder (connected part)
5. Brake pedal	5.1 Brake pedal arm 5.2 Pedal pivot/shaft 5.3 Return spring 5.4 Brake rod or linkage 5.5 Brake light switch (spring/contact type) 5.6 Footrest connection

Curricular Content Guide

1. Underpinning Knowledge	1.1. Concept of braking systems is understood. 1.2. Identify major brake components. 1.3. Understand the function of brake pads and discs in generating friction to slow down the motorcycle. 1.4. Interpret manufacturer’s specifications for brake pad thickness, disc condition, and brake fluid type. 1.5. Understand common brake faults. 1.6. Concept of bleeding brakes is explained. 1.7. Understand safety precautions when handling brake fluid, pads, and calipers to prevent accidents. 1.8. Identify proper alignment of brake pads and calipers to ensure smooth and balanced braking. 1.9. Interpret the role of brake hoses and lines in transmitting hydraulic pressure without leaks. 1.10. Understand the difference between front and rear braking functions for balance and stability.
2. Underpinning Skills	2.1. Inspecting brake pads for thickness, cracks, and uneven wear. 2.2. Checking brake fluid level and condition. 2.3. Inspecting brake lines and hoses for leaks, cracks, or looseness. 2.4. Checking, cleaning, and resetting caliper pistons. 2.5. Reinstalling and tightening brake calipers to specified torque. 2.6. Operating brake lever/pedal to confirm proper pad contact. 2.7. Draining and refilling brake fluid, and bleeding brake lines to remove air. 2.8. Testing brake lever/pedal free play and hydraulic pressure. 2.9. Checking for fluid leaks and cleaning after service. 2.10. Performing controlled road tests to ensure braking efficiency.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities.

	4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 inspected brake pads, discs, and fluid levels 1.2 replaced worn-out brake pads and adjust callipers 1.3 performed bleed the brake lines and replace brake fluid 1.4 tested the brakes
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **MAINTAIN ELECTRICAL SYSTEM OF A MOTORCYCLE**	**Nominal Duration:** 40 Hrs**.**	**Unit Code:** SICIP-TRA-MCMS-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to maintain electrical system of a motorcycle. It specifically includes the tasks of testing the battery voltage and charging capacity, inspecting wiring, checking the ignition system, replacing faulty fuses and components and verifying the electrical system.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Test the battery voltage and charging capacity	1.1 **Battery terminals** are inspected for corrosion, looseness, or damage. 1.2 Battery voltage is measured with a multimeter and compared to the specified range. 1.3 Engine is started and charging voltage is measured across the terminals. 1.4 Alternator/charging system output is tested at different engine speeds. 1.5 Any undercharging or overcharging condition is identified and noted.
2. Inspect wiring	2.1 Wiring harnesses are visually inspected for cuts, burns, or worn insulation. 2.2 Connectors, terminals, and joints are checked for looseness or corrosion. 2.3 Continuity of wires is tested using a multimeter to detect breaks or faults. 2.4 Signs of overheating, short circuit, or melted insulation are identified. 2.5 Routing of wires is inspected to ensure they are properly clipped and not rubbing against sharp edges.
3. Check the ignition system.	3.1 Ignition switch and key operation are checked for proper function. 3.2 Spark plug condition is inspected for wear, fouling, gap, and damage. 3.3 Spark plug cable, ignition coil, and Capacitor discharge ignition (CDI)/Electronic Control Unit (ECU) connections are examined for looseness or corrosion. 3.4 Spark quality is tested by cranking the engine and observing consistent spark. 3.5 Ignition timing and firing sequence are checked against manufacturer's specifications.
4. Replace faulty fuses and components.	4.1 Fuse box is opened and fuses are inspected for burns, breaks, or discoloration.

	4.2 Faulty fuses are removed and replaced with new ones of the correct rating. 4.3 **Electrical components** are tested for functionality. 4.4 Damaged or faulty components are removed using proper tools. 4.5 New components are installed according to manufacturer's specifications. 4.6 Electrical system is tested to confirm proper operation after replacement.
5. Verify electrical system.	5.1 Ignition system, switches, and controls are checked for correct operation. 5.2 **All lighting units** (are tested for brightness and function. 5.3 Horn, meters, and warning lights are checked for accuracy and soundness. 5.4 Battery voltage and charging output are re-tested to ensure stable performance. 5.5 Electrical continuity and grounding are verified with a multimeter.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Battery terminals	1.1 Post Terminals (Top Post) 1.2 Side Post Terminals 1.3 Screw Terminals 1.4 L-Type Terminals
2. Electrical components	2.1 Bulbs 2.2 Switches 2.3 Relays 2.4 Sensors
3. All lighting units	3.1 Headlight 3.2 Tail light 3.3 Indicators 3.4 Brake light

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify the function of the battery as the main power storage unit and source of electrical energy. 1.2 Understand charging system components and their role in maintaining battery health. 1.3 Interpret battery voltage readings using a multimeter. 1.4 Identify signs of corrosion or damage on battery terminals and connectors that affect conductivity. 1.5 Understand the role of wiring harnesses in distributing power safely across the motorcycle. 1.6 Identify common wiring faults. 1.7 Interpret continuity tests with a multimeter.

	1.8 Understand the purpose of fuses and their correct ratings to protect electrical components. 1.9 Identify the role of the ignition system in generating and timing the spark. 1.10 Interpret spark plug condition as indicators of engine performance. 1.11 Understand safe handling procedures for batteries, wiring, and electronic parts to prevent accidents. 1.12 Interpret final testing results to confirm system reliability.
2. Underpinning Skills	2.1 Inspecting battery terminals for corrosion, looseness, or damage. 2.2 Measuring battery voltage and comparing it to specifications. 2.3 Testing charging system output at different engine speeds. 2.4 Identifying and noting undercharging or overcharging conditions. 2.5 Inspecting wiring harnesses, connectors, terminals, and joints for damage or corrosion. 2.6 Testing wire continuity and identifying shorts, breaks, or overheating. 2.7 Testing spark quality and verifying ignition timing and firing sequence. 2.8 Installing new electrical components according to manufacturer's specifications. 2.9 Testing electrical system, lighting, horn, meters, and warning lights for proper operation. 2.10 Verifying electrical continuity and grounding with a multimeter.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 tested the battery voltage and charging capacity.

	1.2 inspected wiring. 1.3 checked the ignition system. 1.4 replaced faulty fuses and components. 1.5 verified the Electrical system.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	**Motor Cycle Maintenance and Servicing**
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sq) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computers/ Laptop	01
2.	Multimedia Projector	01
3.	Internet connectivity	01
4.	Motorcycles	05
5.	Motorcycle lift/stand	02
6.	Chain breaker and riveter tool	02
7.	Bearing puller set	02
8.	Avometer/multimeter	02
9.	Battery charger/maintainer	01
10.	Universal Tire Air Pressure Gauge Dial Meter Tester	02
11.	Tire changing tools (bead breaker, tire levers, balancing stand)	02
12.	Wheel balancer	01
13.	Air compressor	01
14.	Tool box (Ball Peen Hammer, Files, Allen Key, Open And Ring Wrenches, Wrenches, Socket Wrench Set, Pliers, Puller, Scriber, Filler Gauge, Plug Wrench, Tap Sets, Die Sets, Hacksaw, Spanners, Vice Grip, Wire Cutters, And Clamps)	05
15.	Power tools (Drills, Grinders and Soldering iron)	05

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Motor Cycle Maintenance and Servicing the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.